BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 106 · Algorithm Design and Flowcharts

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is abstraction?

Q2. What does WHERE do in SQL?
A. Names the source table
B. Filters rows
C. Chooses output columns
D. Sorts all rows

Q3. A zero-indexed 2D array is [[11,12],[21,22]]. Give grid[1,0], then state the indexes needed to access 12.

Q4. Two mono recordings have the same sample rate and bit depth. One lasts twice as long. Does this double its quality? How does raw size change?

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I Do · Read new knowledge

Key vocabulary

Flowchart — Diagram representing algorithm steps and control flow.

Decision — Diamond that selects between paths using a condition.

Process — Rectangle representing an operation.

Input/output — Parallelogram for reading or displaying data.

Iteration — Repeating a set of instructions.

New learning

Clear algorithms: An algorithm must be finite, precise and able to produce an intended result for allowed inputs.

Basic flowchart shapes: Use terminal (start/end), process (rectangle), input/output (parallelogram), decision (diamond) and labelled arrows.

Selection and iteration: A decision directs execution along yes/no paths; an invalid input can loop back to the input step.

Equivalent pseudocode: Trace both representations on the same input to check they perform the same actions.

Worked example

Design a flowchart that repeatedly asks for a mark until it is between 0 and 100 inclusive. [5 marks]

1. Establish the input

2. START → input mark using a parallelogram.

3. Decision checks whether 0 ≤ mark ≤ 100; YES continues to END.

4. NO returns to input mark, rather than ending; label both arrows. [5]

We Do · Practise with guidance

Guided 1

Write the steps for an algorithm that asks for an age until age ≥ 0. [4]

Worked method: Start/input age [1]; decision age ≥ 0 [1]; valid branch ends/continues [1]; invalid branch loops to input [1].

Guided 2

Explain the difference between a process symbol and decision symbol. [4]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Removing irrelevant detail while keeping what is needed to solve the problem.

Q2. B — Filters rows

Q3. grid[1,0] is 21; 12 is at grid[0,1].

Q4. Raw size doubles: there are twice as many samples. Quality does not improve. Sample rate and bit depth stay the same.

I Do / We Do pages · Guided practice

Q1. Start/input age [1]; decision age ≥ 0 [1]; valid branch ends/continues [1]; invalid branch loops to input [1].

Q2. Process is a rectangle [1] representing an operation/calculation [1]; decision is a diamond [1] selecting a path depending on a condition [1].

Q3. -1 is rejected as below range [1]; arrow loops back [1]; 75 is within 0–100 [1]; flow follows valid branch to end [1].

You Do pages · Code First

```
octet = 260
if octet >= 0 AND octet <= 255 then
    print("Valid range")
else
    print("Invalid")
endif
```

Expected result: Invalid

You Do pages · Independent questions

Q1. Diamond [1].

Q2. 0 fails validity [1] and loops to input [1]; 6 passes [1] and algorithm continues/exits [1].

Q3. Records changes in variables [1] for each relevant step [1] to identify unexpected logic/errors [1].

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Worked method: Process is a rectangle [1] representing an operation/calculation [1]; decision is a diamond [1] selecting a path depending on a condition [1].

Guided 3

Trace a validation flowchart when marks entered are -1, 75. [4]

Worked method: -1 is rejected as below range [1]; arrow loops back [1]; 75 is within 0–100 [1]; flow follows valid branch to end [1].

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Correct the Boolean operator used for checking an IPv4 octet.

```
octet = 260
if octet >= 0 OR octet <= 255 then
    print("Valid range")
else
    print("Invalid")
endif
```

iterate

Question 1

State the shape normally used for a decision in a flowchart. [1] Source: Original OCR J277-style practice question.

Question 2

An algorithm must accept an integer from 1 to 10. Describe the paths for 0 followed by 6. [4] Source: Original OCR J277-style practice question.

Question 3

Explain one reason for using a trace table while refining an algorithm. [3] Source: Original OCR J277-style practice question.

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Exit Ticket · Answer before checking

A trace table can help check how values change during a program.